



Car Price Prediction Using Machine Learning

Submitted By

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Project Title: Car Price Prediction Using Machine Learning

Technology Used: Python, Machine Learning

Algorithm Used: Linear Regression

Abstract

This project focuses on predicting the selling price of used cars using Machine Learning. The model analyzes various features such as the present price, age of the car, kilometers driven, fuel type, seller type, transmission type, and ownership history to estimate the expected selling price. The project follows the complete machine learning workflow, including data preprocessing, exploratory data analysis (EDA), feature engineering, model training, evaluation, and prediction. The developed model can assist buyers, sellers, and dealerships in estimating a fair resale value for used vehicles.

1. Introduction

The resale value of a car depends on several factors, including its age, condition, fuel type, transmission, and current market value. Accurately estimating a car's selling price helps buyers make informed purchasing decisions and enables sellers to price their vehicles competitively.

Machine Learning provides an efficient way to analyze historical data and identify patterns that influence car prices. In this project, a Linear Regression model is trained to predict the selling price of used cars based on their features.

2. Objectives

- Predict the selling price of used cars.
 - Clean and preprocess the dataset.
 - Perform Exploratory Data Analysis (EDA).
 - Engineer new features for improved prediction.
 - Encode categorical variables.
 - Train a Linear Regression model.
 - Evaluate the model using regression metrics.
 - Generate business insights from the analysis.
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3. Tools and Technologies

Tool	Purpose
Python	Programming Language

Tool	Purpose
Jupyter Notebook	Development Environment
Pandas	Data Manipulation
NumPy	Numerical Computation
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Scikit-learn	Machine Learning

4. Dataset Description

The dataset contains information about used cars and their selling prices.

Feature	Description
Car_Name	Name of the car
Year	Manufacturing year
Selling_Price	Selling price (Target Variable)
Present_Price	Current showroom price
Driven_kms	Distance driven by the car

Feature	Description
Fuel_Type	Petrol / Diesel / CNG
Selling_type	Dealer or Individual
Transmission	Manual or Automatic
Owner	Number of previous owners
Total Records: 301	

5. Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked for missing values.
- Checked for duplicate records.
- Dropped the **Car_Name** column as it is not directly useful for prediction.
- Created a new feature called **Car_Age** from the manufacturing year.

- Removed the **Year** column after creating **Car_Age**.
 - Converted categorical variables into numerical format using One-Hot Encoding.
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6. Exploratory Data Analysis (EDA)

Several visualizations were created to understand the dataset.

Visualizations

- Correlation Heatmap
 - Selling Price Distribution
 - Present Price vs Selling Price
 - Driven Kilometers vs Selling Price
 - Actual vs Predicted Price
 - Residual Plot
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7. Feature Engineering

A new feature, **Car_Age**, was created using:

$\text{Car_Age} = \text{Current Year} - \text{Manufacturing Year}$

Categorical columns were converted into numerical values using One-Hot Encoding:

- Fuel_Type
 - Selling_type
 - Transmission
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8. Machine Learning Model

The project uses the **Linear Regression** algorithm.

Input Features

- Present Price
- Driven Kilometers
- Owner
- Car Age
- Fuel Type
- Seller Type
- Transmission

Target Variable

- Selling Price
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9. Model Training

The dataset was divided into:

- **Training Data:** 80%
- **Testing Data:** 20%

The Linear Regression model was trained using the training dataset and tested on the remaining data.

10. Model Evaluation

The model was evaluated using the following metrics:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Metric Value

MAE *(Your Output)*

MSE *(Your Output)*

RMSE *(Your Output)*

R^2 Score *(Your Output)*

11. Business Insights

- Cars with a higher present showroom price generally have a higher resale value.
 - Newer cars tend to retain more value than older cars.
 - Cars with fewer kilometers driven generally have a better resale price.
 - Fuel type and transmission influence market demand and resale value.
 - The model can assist dealerships and customers in estimating a fair selling price.
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12. Conclusion

This project successfully demonstrates the use of Machine Learning for predicting used car prices. The Linear Regression model provides reliable price estimates based on multiple vehicle characteristics. The project highlights the importance of data preprocessing, feature engineering, and model evaluation in building effective predictive models.

13. Future Scope

- Compare Linear Regression with Random Forest Regressor, Decision Tree, and XGBoost.
 - Perform hyperparameter tuning.
 - Deploy the model as a Streamlit web application.
 - Save and load the trained model using Joblib.
 - Train on larger and more diverse datasets for improved accuracy.
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14. References

1. Python Documentation
 2. Pandas Documentation
 3. NumPy Documentation
 4. Matplotlib Documentation
 5. Seaborn Documentation
 6. Scikit-learn Documentation
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15. Appendix

Include the following:

- Sample dataset (car_data.csv)
- Screenshots of the Jupyter Notebook
- Model evaluation output
- Prediction example
- Graphs from the images/ folder